

Math Word Problem Solving by Generating Linguistic Variants of Problem Statements

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Introduction

Math Word Problems (MWPs)

Definition:
A Math Word Problem is a textual narrative that states a problem description and poses a question about one or more unknown quantities. (found in 1st to 8th grade math text-books)

Example: 69 handbags are sold for $23 each. There are a total of 420 handbags in a boutique and the remaining handbags are sold for $17 each. How much did the boutique earn after selling all the handbags?
Expression: $X = 69 \times 23 + (420 - 69) \times 17$

Goal:
Translate a MWP statement to a **valid** math expression, which when evaluated, yields the **solution** to the problem.

Objectives

Math Word Problem Solvers

- Automated systems with NLP models to solve MWPs, given the problem statement in Natural Language, with reasonable accuracy.

- Focus on addressing the challenges of:

  - ✓ Understanding the quantities in the problem and capturing their complex mathematical interconnections from a linear textual sequence, which can sometimes be ambiguous.

  - ✓ Diverse range of MWPs with differing difficulty levels, i.e., varying number of unknown values and depth of the relationships between quantities.

  - ✓ Absence of crucial information and presence of irrelevant information in the problem statements.

  - ✓ Chronological and temporal ambiguities of the events happening in the problem statements.

  - ✓ MWPs that significantly differ from the training set in terms of semantic and syntactic structure.

Motivation

Our Inspiration to Pursue the Topic

- Emulating the mathematical aptitude of humans in machines — crucial frontier of research towards general AI.
- Utility — Smart Education Tool for teaching Math.
  - One of the talking points in a recent Neural Information Processing Systems (NeurIPS) conference titled *"Math AI for Education (MATHAI4ED): Bridging the Gap Between Research and Smart Education"*
- The International Math Olympiad (IMO) Grand Challenge [1].
  - To create AI models capable of solving Math Olympiad problems.
- Incentive from being avid problem-solvers ourselves! — we hope to try our best to work and contribute to this domain.

Problem Formulation

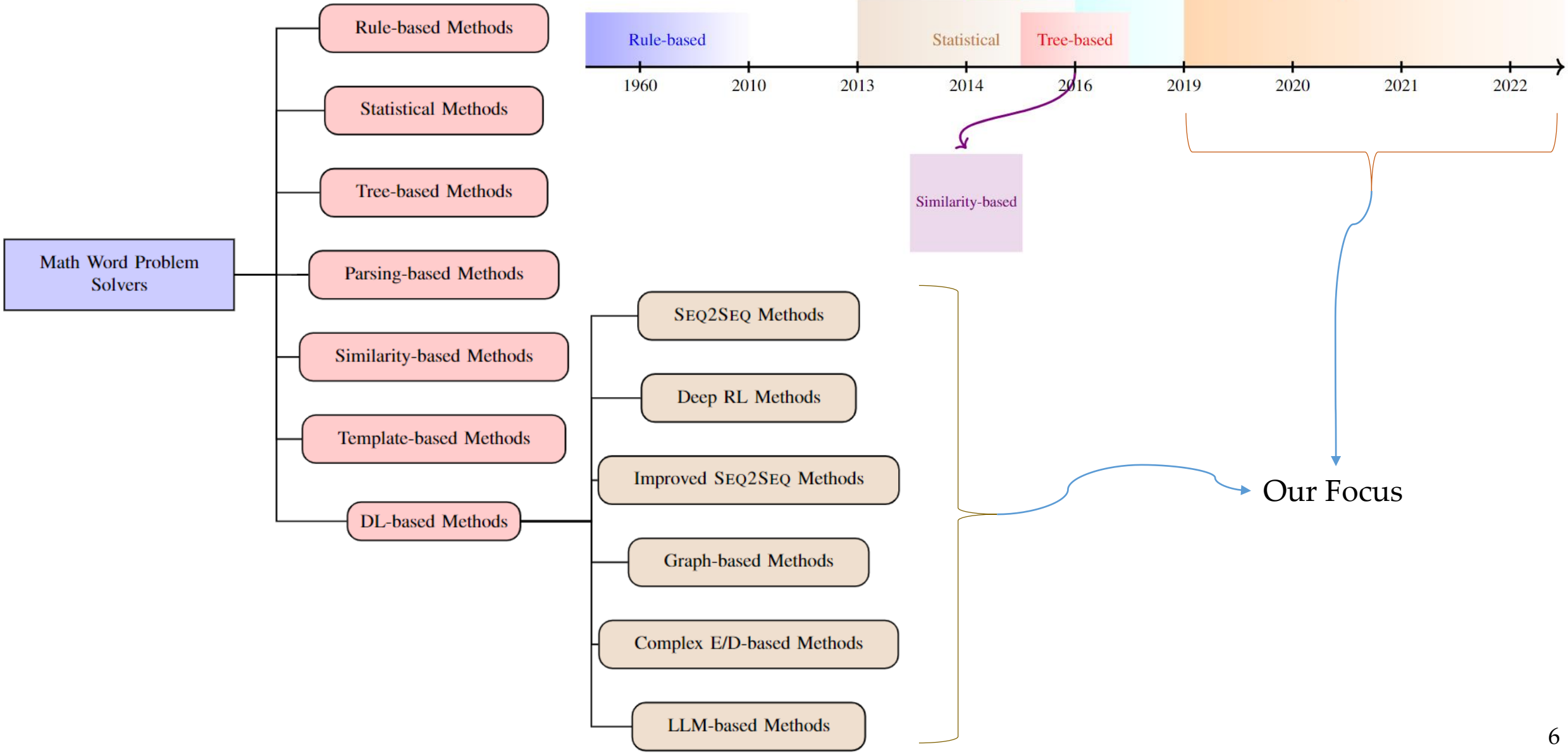
The Precise Statement of the Problem

A Math Word Problem $\boldsymbol{P}$ is a sequence of word tokens and numeric values, where the $\boldsymbol{V_P} = \{v_1, \ldots, v_m\}$ denotes the word tokens in $\boldsymbol{P}$ and the set $\boldsymbol{n_P} = \{n_1, \ldots, n_l\}$ denotes the set of numeric quantities in $\boldsymbol{P}$. The set of word tokens $\boldsymbol{V_P}$ consists of entities such as names of people, objects, units and rates while the set of quantities $\boldsymbol{n_P}$ consists of the numerical amount relevant to those entities.

The goal of an MWP solver model is to map $\boldsymbol{P}$ to a valid mathematical expression $\boldsymbol{E_P}$, consisting of the quantities in $(\boldsymbol{n_P} \cup \boldsymbol{C})$, where $\boldsymbol{C}$ is a set of constants, and the fundamental mathematical operators $\boldsymbol{O} = \{+, -, \times, \div\}$, which can be evaluated to obtain the correct answer.

Literature Review

Taxonomy and Chronology



Literature Review

Rule-based Methods [2, 3, 4]

- Rule based methods are chronologically some of the earliest approaches of solving MWPs. They split the problem text to a set of propositions and apply algorithms with hard-coded rules to obtain the expressions.

Prominent Works:

- WORDPRO [2] — coded in Interlisp-D, which could solve one-step arithmetic problems with 4 types of predefined schemata after translating the problem statement to a set of propositions.

- ROBUST [3] — which could conceptualize free-format multi-step MWPs with extraneous information by introducing 6 types of predefined schemata.

- MSWPAS [4] — which could solve multi-step addition and subtraction problems by converting the problem statements into Problem Frames consisting of the whole semantic information of the problems.

Disadvantages:

- High **dependency** on manual features
- **Inability** to generate novel templates for new problems

Literature Review

Statistical Methods [5, 6, 7, 8, 9, 10, 11]

- The Statistical methods use generic ML models to extract the entities, quantities and operators from the problem statement and infer the numeric answer with simple logic.

Prominent Works:

- (Kushman *et al.*)[5] - reason across sentence boundaries and define a joint log-linear distribution over systems of equations including the interrelations between those equations and the problem text.
- (Roy *et al.*)[6] – Quantity Entailment scheme, solves word problems with a single operator, uses a cascade of 3 classifiers.
- (Hosseini *et al.*)[7] - ARIS — first to introduce a 7-category Verb Categorization and synsets for predicting verb categories in sentences using syntactic and semantic features.
- (Liang *et al.*)[10, 11] - analyze the problem statement and transform both the problem scenario and the question at the end of the problem statement into their tag-based logic forms and eventually infer the solution.

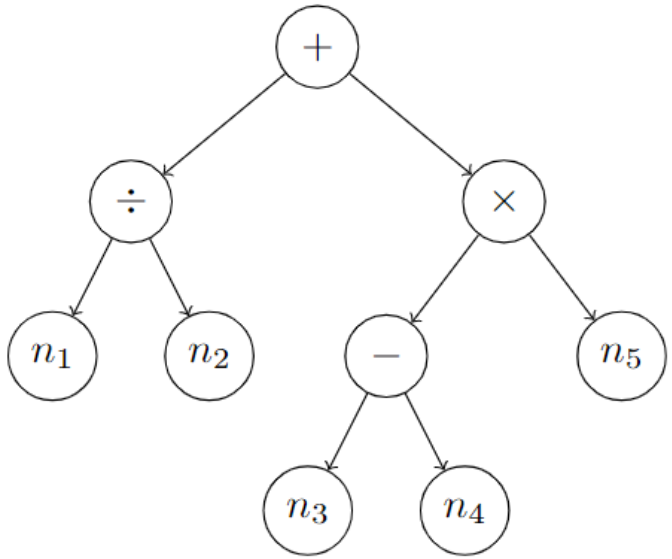
Disadvantages:

- They **need additional manual annotation** which is **time-consuming** and makes them **unsuitable for large data sets**.
- They heavily **rely on a set of pre-defined templates** which are not flexible. Expanding these templates into the realm of multiplication (×) and division (÷) will **require tremendous manual effort**.

Literature Review

Tree-based Methods [12, 13, 14, 15, 16, 17, 18]

- Tree-based methods of solving MWPs focus on constructing the structure of the resultant expression/equation's equivalent tree in a bottom-up manner.



Expression Tree representing $n_1 \div n_2 + (n_3 - n_4) \times n_5$.

Prominent Works:

- (Roy *et al.*) [12] – binary SVM classifier for quantities, Lowest Common Ancestor (LCA) prediction for each pair of leaf nodes.
- (Koncel-Kedziorski *et al.*) [14] – ALGES — brute-force generation of all possible equation trees using Integer Linear Programming (ILP).

Advantages:

- Additional manual annotations *e.g.* template, tags or logic forms are not required.

Disadvantages:

- Additional **computational overhead.**

Literature Review

Parsing-based Methods [19, 20]

- Leverages a joint representation to automatically learn the association between the words in the problem text and math expressions which possesses semantic *"closeness"*.

Prominent Works:

- (Shi *et al.*) [19] – SIGMADOLPHIN — Uses their created Dolphin Language (DOL) as representation language to convert MWP to DOL trees.

- (Zou *et al.*) [20] – TEXT2MATH — novel joint representation to automatically learn the association between the words in the problem text and math expressions

Disadvantages:

- Their **applicability is limited** to Number Word Problems. (*e.g.* What is 2 times 3?)
- They **show incompetence** in **grasping the complex relationships** between quantities and entities seen on general MWPs.

Literature Review

Similarity-based Methods [21]

- Works by calculating the similarity between a test set sample with samples in the training set.

Prominent Work:

- (Huang *et al.*) [21] – SIM — Equation system of the most similar training sample (template selection) is applied to the test sample problem under consideration (template slot filling).
  - ➢ Uses Term Frequency-Inverse Document Frequency (TF-IDF) and Jaccard Similarity scores.

Disadvantages:

- They **fail to work out problems** that do not necessarily follow the same structural template of the problems in the training set.

Literature Review

Template-based Methods [5, 8, 15, 22, 23]

- The core idea behind template-based approaches is to identify a candidate template from a pre-defined corpus of equation templates and plug in the numeric and variable slots with quantities extracted from the problem statement.

Prominent Works:

- (Upadhyay *et al.*) [22] – MIXEDSP — learns from both explicit supervision (e.g. equations) and implicit supervision (e.g. solutions) based in Structured-output Perceptron
- (Huang *et al.*) [23] – FGEXPRESSION - they defined a template fragment as any subtree with at least a single operator and 2 operands. The mapping procedure was *"fine grained"* based on Longest Common Substring (LCS).

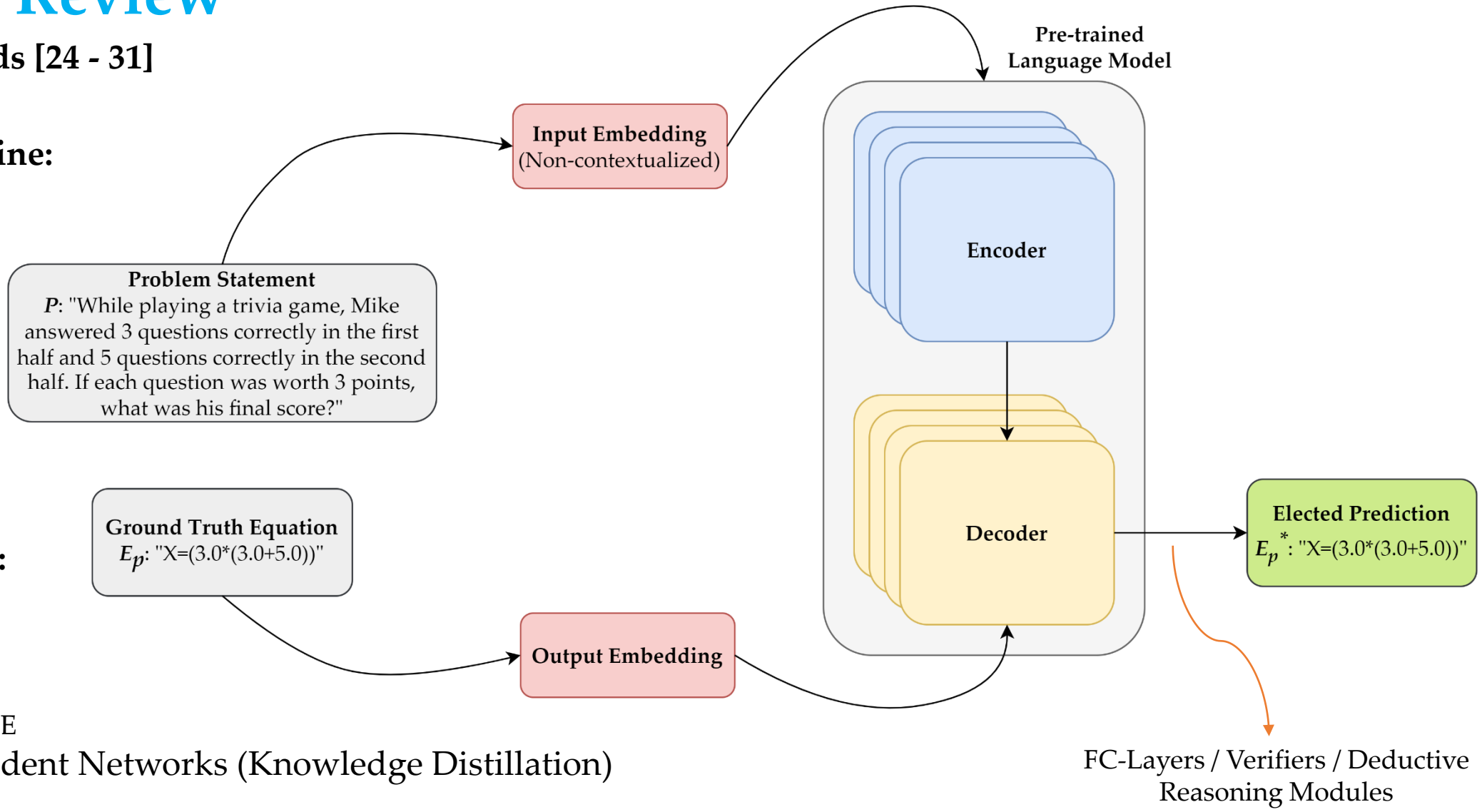
Disadvantages:

- They **fail to work** well in sparse training samples.
- The learning process **heavily relies** on lexical and syntactic features with huge and sparse feature space, *e.g.*, the dependency edge/path between two slots of a template.

Literature Review

DL-based Methods [24 - 31]

General Pipeline:



Subcategories:

- SEQ2SEQ
- Deep RL
- SEQ2TREE
- GRAPH2TREE
- Teacher-Student Networks (Knowledge Distillation)
- BERT2TREE
- BART2TREE
- ROBERTA2TREE
- LLM (few-shot prompt-based, no weight updates)

Literature Review

A Goal-Driven Tree-Structured Neural Model for Math Word Problems [25]

Xie, Zhipeng, and Shichao Sun, IJCAI, 2019.

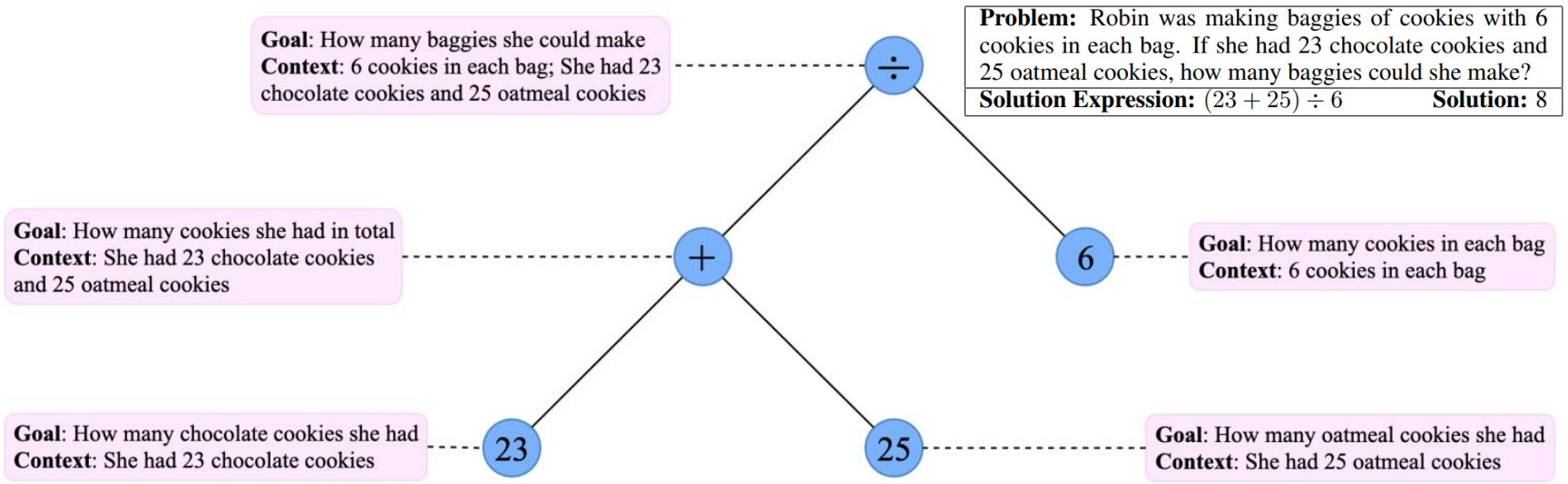
This paper proposes a **tree-structured** neural model to generate expression tree in a **goal-driven** manner. Given a math word problem, the model first identifies and encodes its goal to achieve, and then the goal gets decomposed into sub-goals combined by an operator in a top-down recursive way.

Contributions:

- The **pioneers** in **tree-structured** neural model for MWPs.
- In their model, GTS, problem text information explicitly flows through the expression tree in both **top-down** (*goal decomposition*) and **bottom-up** (*subtree embedding*) manners.
- Significantly outperformed the then several state-of-the-art (SOTA) systems on the Mandarin Chinese MWP dataset MATH23K [26].

Drawbacks:

- It doesn't recognize various mathematical laws such as Commutative Law in Addition and Multiplication.
- It fails to generate more than one valid solution expression for a problem.



Literature Review

Graph-to-Tree Learning for Solving Math Word Problems [27]

J. Zhang, L. Wang, R. K.-W. Lee, Y. Bin, Y. Wang, J. Shao, and E.-P. Lim, Association for Computational Linguistics (ACL), 2020.

GRAPH2TREE — A novel deep learning architecture that combines the merits of the **graph-based encoder** and **tree-based decoder** to generate better solution expressions.

Contributions:

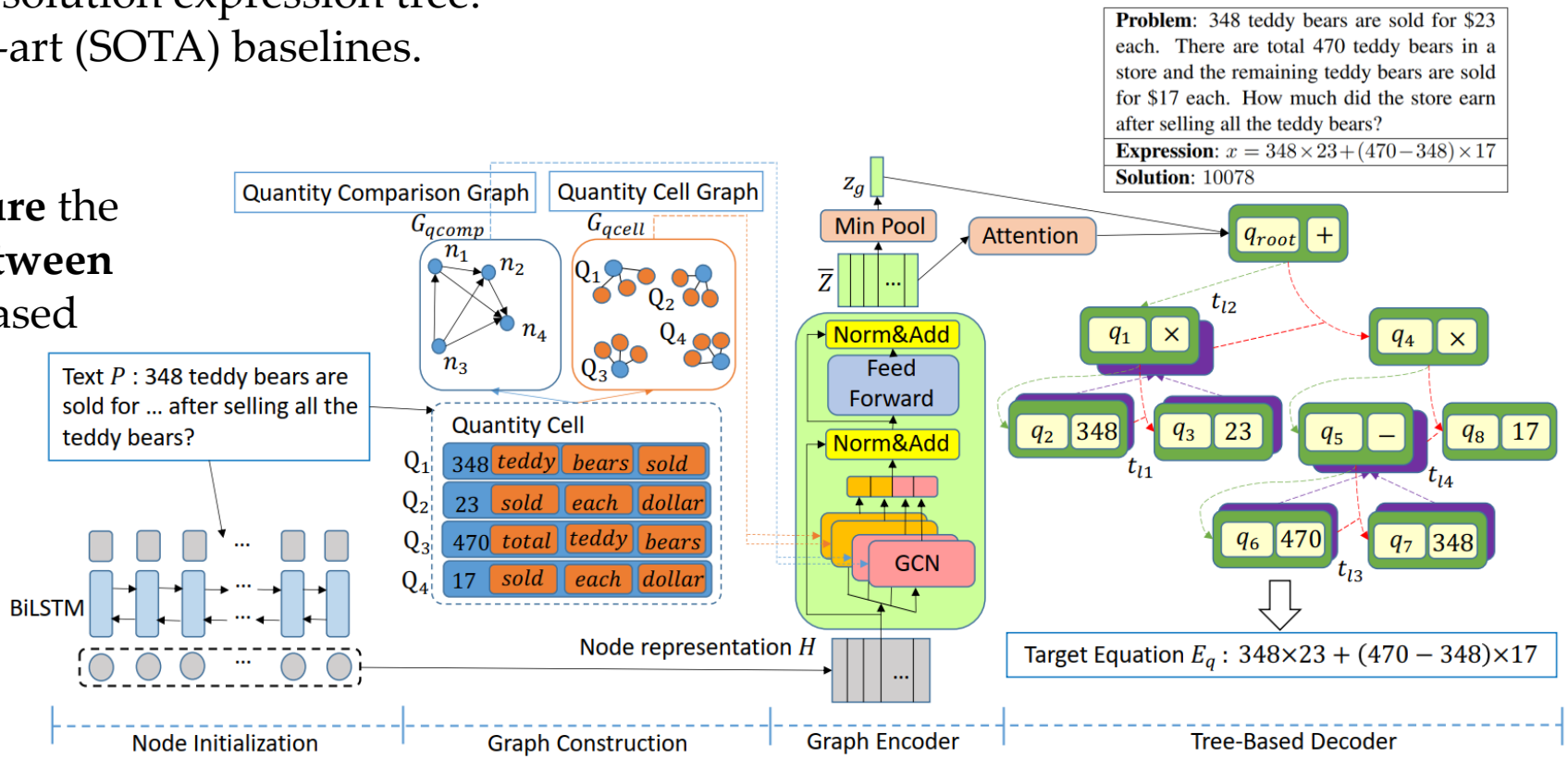
- Construction of the **Quantity Cell Graph** and **Quantity Comparison Graph**.
- A **graph transformer** to learn the latent quantity representations from their proposed graphs, and a **tree-structure decoder** to generate a solution expression tree.
- Outperformed then state-of-the-art (SOTA) baselines.

Advantages:

- Graph-based Encoder can **capture** the **relationships and associations between quantities better** than Sequence-based Encoder.

Drawbacks:

- It still **falters in the cases of long solution expressions** and **problems that require complex reasoning** to solve.

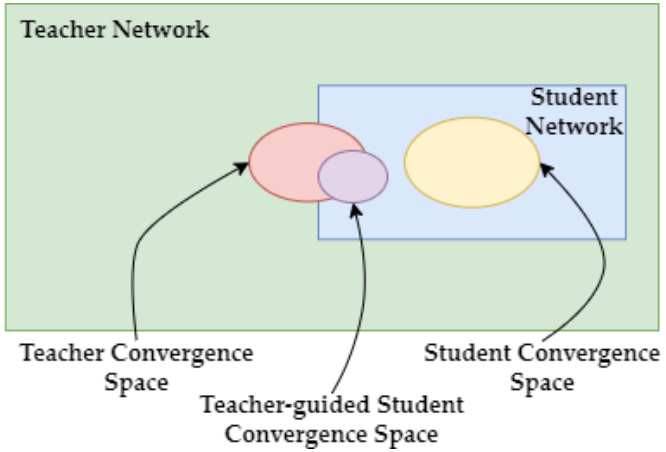


Literature Review

Teacher-Student Networks with Multiple Decoders for Solving Math Word Problem [28]

J. Zhang, R. K.-W. Lee, E.-P. Lim, W. Qin, L. Wang, J. Shao, and Q. Sun, IJCAI, 2020.

TSN-MD — leverages a teacher network to integrate the knowledge of equivalent solution expressions such as to better regularize the learning behavior of the student network. (**Knowledge Distillation**)



Contributions:

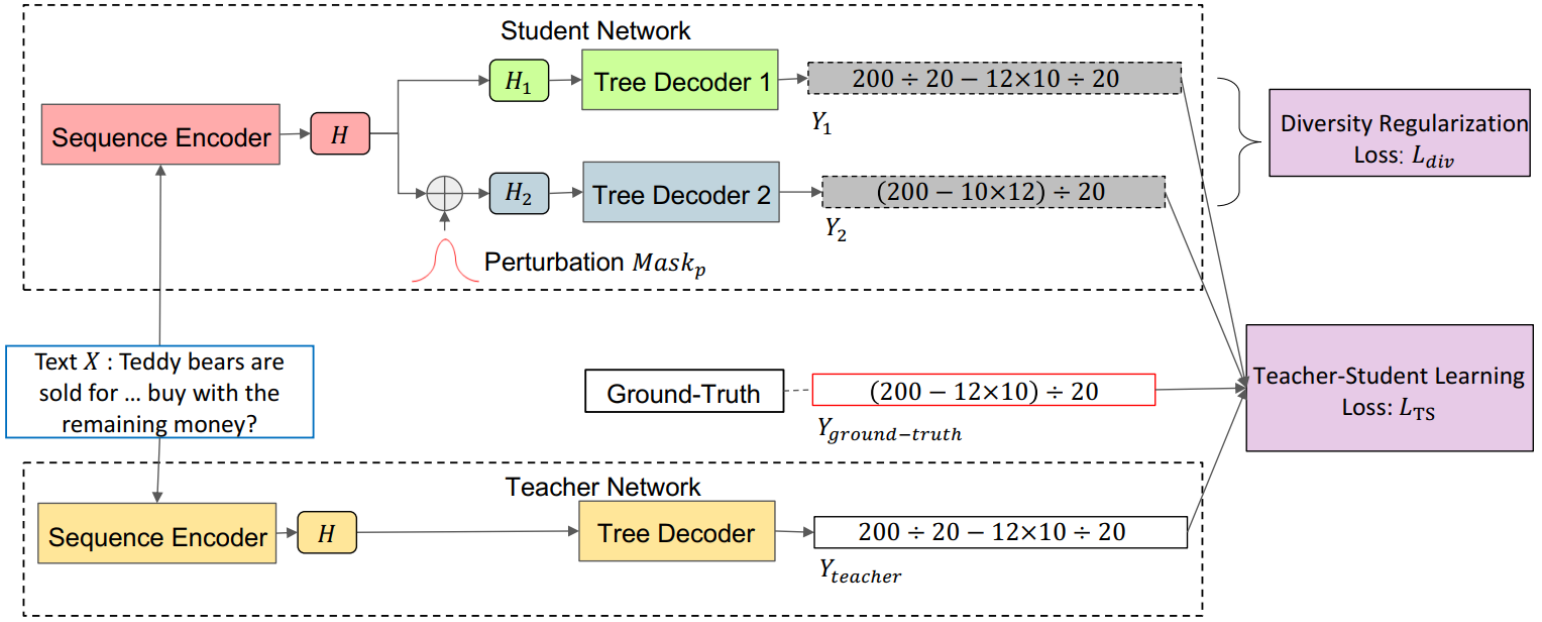
- A **multiple-decoder** architecture that works for our **teacher-student networks** to improve the **diversity** of generated expressions.
- **Superiority** over the other then MWP solving approaches on MAWPS and MATH23K datasets.

Advantages:

- Can produce novel expressions whose templates are not present in the original dataset.

Drawbacks:

- The model doesn't penalize the generation of such novel expressions if they don't end up yielding the correct final answer.



Literature Review

MWP-BERT: A Strong Baseline for Math Word Problems [29]

Z. Liang, J. Zhang, J. Shao, and X. Zhang, 2021. (recent version in Findings of the Association for Computational Linguistics: NAACL 2022)

First to use pre-trained token representations that capture the alignment between text description and mathematical logic using the pre-trained **Bidirectional Encoder Representations from Transformers (BERT)** language model.

Contributions:

- Used **BERT-encoder** and **Tree-decoder**.
- For problems that need **commonsense knowledge**, they used a **keyword-based prompt matching** method.
- Curated a new dataset called APE-CLEAN from APE210K.
- Outperformed then baselines on MATH23K and APE210K.

Advantages:

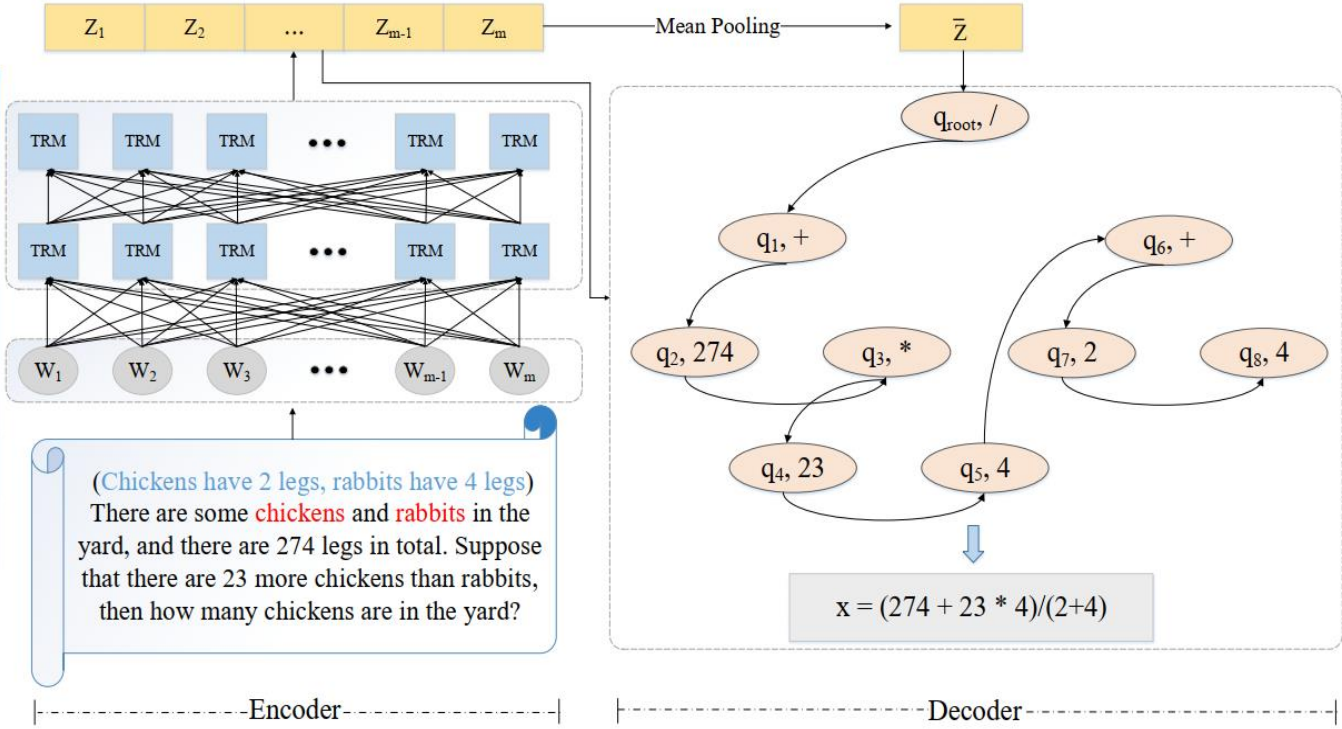
- Utilizes the good Natural Language Understanding (NLU) ability of BERT.

Drawbacks:

- **Fails** to solve inverse versions of problems (Patel *et al.*) [24].
- **Not sufficiently adept** at modelling either language or math [24].
- Offline table **not scalable** to English problems.

Keyword	Sentence
Chicken	Chickens have 2 legs
...	.....
...	.....
Rabbit	Rabbits have 4 legs
...	.....
...	.....

Keyword-sentence Prompts



Literature Review

Improving Math Word Problems with Pre-trained Knowledge and Hierarchical Reasoning [30]

W. Yu, Y. Wen, F. Zheng, and N. Xiao, in Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2021.

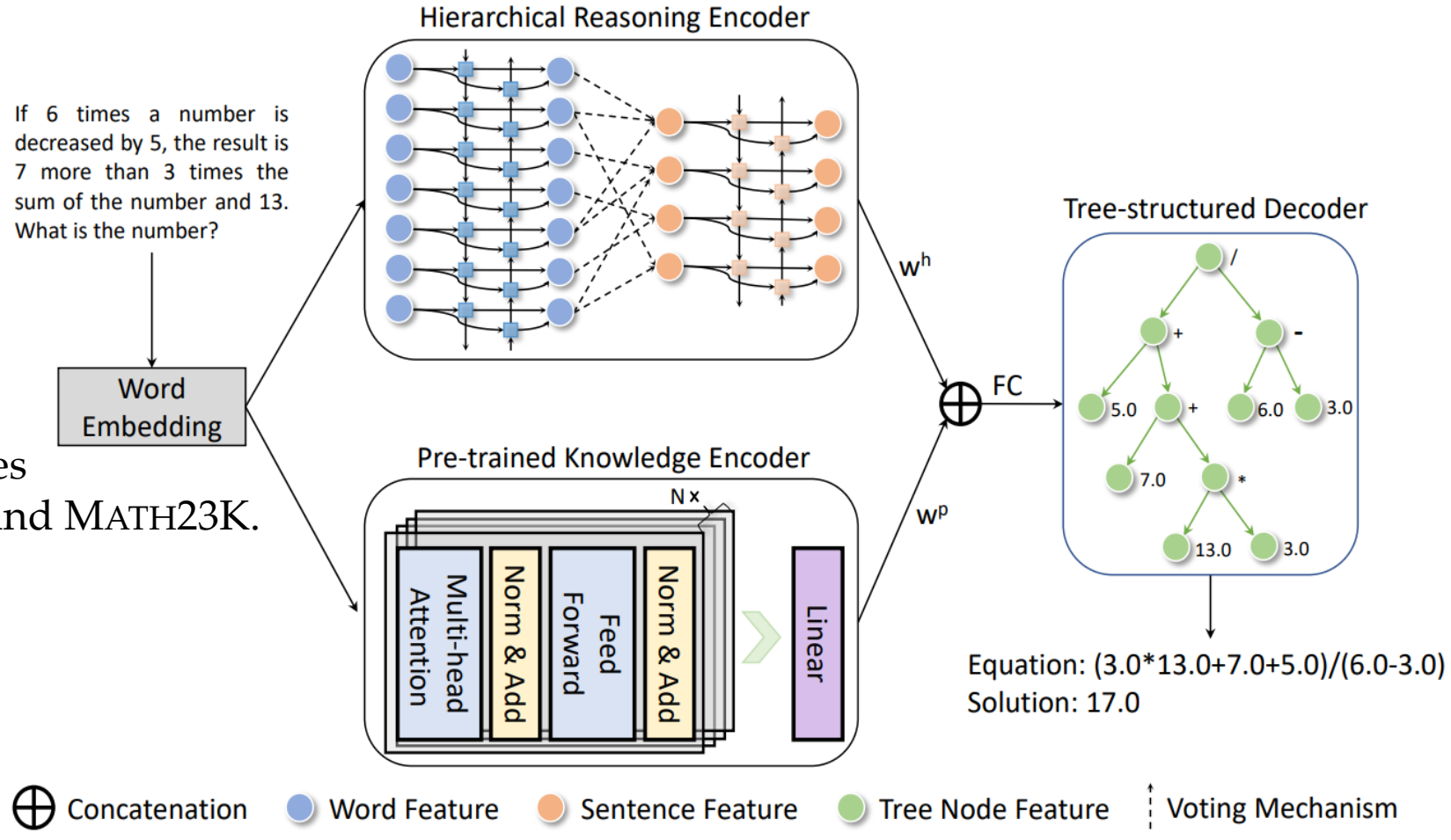
RPKHS — Bridges the hierarchical reasoning between the entity (word-level) and context (sentence-level) unlike previous works which mainly focus on designing diverse entity-level structures for word-level reasoning.

Contributions:

- Pioneers to explore pre-trained knowledge on the MWP task.
- A hierarchical reasoning encoder to integrate word-level & sentence-level reasoning for bridging the entity and context domain.
- A Pre-trained knowledge encoder with BERT and ROBERTA.
- Outperformed previous approaches by a significant margin on MAWPS and MATH23K.

Advantages:

- Concatenation of semantics between entity and context and knowledge points from the linguistic world (filtering irrelevancy).



Literature Review

Learning to Reason Deductively: Math Word Problem Solving as Complex Relation Extraction [31]

Z. Jie, J. Li, and W. Lu, 2022.

RoBERTa-DeductReasoner — a deductive reasoning module at downstream of the model which yields human-understandable deductive reasoning steps iteratively and constructs the final output expression.

Contributions:

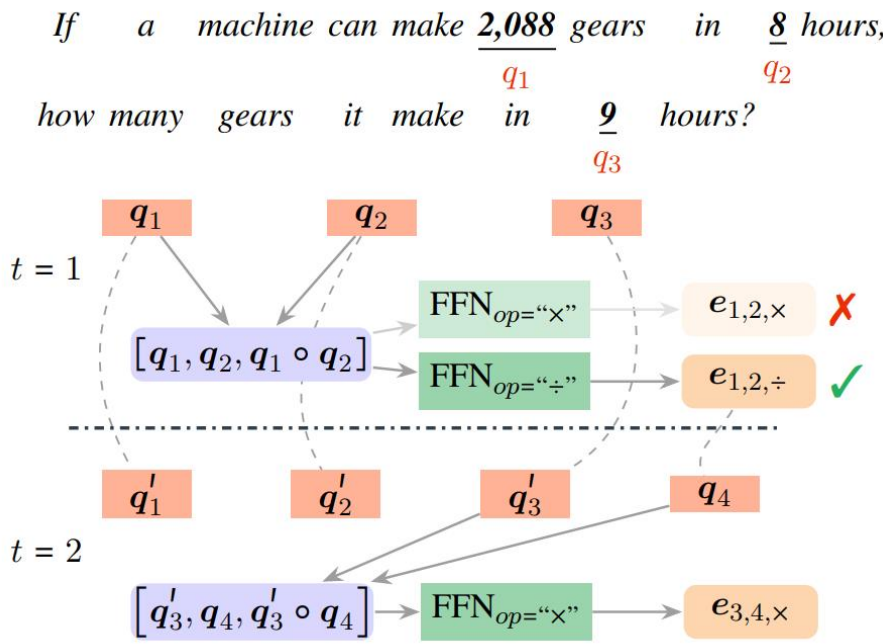
- Newly defines MWP solving as a **complex relation extraction** task.
- Model is able to automatically produce explainable steps that lead to the final answer.
- Significantly outperforms existing strong baselines and currently SOTA non-gigantic PLM on the English MWP datasets MAWPS (**92.0%**) and SVAMP (**47.3%**).

Advantages:

- With deductive reasoning, make **correct equation predictions** on problem statements that require **somewhat complex line of reasoning**.

Disadvantages:

- Susceptible to **failure** in some adversarial problem statements.
- Hard-coded constant quantity list.



Question: *There are **255** apple trees in the orchard. Planting another **35** pear trees makes the number exactly the same as the apple trees. If every **20** pear trees are planted in a row, how many rows can be planted in total?*

Gold Expr: $(255 - 35) \div 20$ **Answer**: 11

Predicted Expr: $(255 + 35) \div 20$ **Predicted**: 14.5

Deductive Scores:

$255 + 35 = 290$ Prob.: 0.068 > $255 - 35 = 220$ Prob.: 0.062

Perturbed Question: *There are **255** apple trees in the orchard. **The number of pear trees is 35 fewer than the apple trees.** If every **20** pear trees are planted in a row, how many rows can be planted in total?*

$255 + 35 = 290$ **Prob.:** 0.061 < $255 - 35 = 220$ **Prob.:** 0.067

Research Challenges

Main Disadvantages of the DL & pre-trained LM-based MWP solvers

- The pre-trained language model-based approaches discussed in the Deep Learning subsection of the Literature Review section **suffer from spurious correlations** between input problem statement and output math expression.

- The existing models,
  - **Depend on shallow heuristics** to yield high degree of accuracy in benchmark datasets, thus **masquerading under the guise of faux mathematical reasoning**.

  - Can even infer solutions to problem statements just by **looking at the superficial patterns** present even if it is lacking word-order information or lacking the actual question text (Patel *et al.*) [24].

  - Prone to failure in the cases of paraphrased, inverse and adversarial variants of problems.

- Absence of **large-scale English** datasets with **adversarial, challenging** and **inverse** versions of MWP samples that require multiple deductive steps and operations to solve and have a good number of **unique math expression templates**.

Research Question

What we are trying to address

- To address the shallow reasoning ability and language understanding,
  - ➢ We propose a **model** to check if we can strengthen the reasoning ability introducing paraphrased, inverse and adversarial variants of problems to the model.

- To address the lack of challenging large-scale English MWP datasets,
  - ➢ We introduce **PARAMAWPS**, a challenging **data augmenting technique** with paraphrased variants and inverse versions of selectively sampled MWPs from the well-known benchmark MAWPS [32] dataset.

Proposed Model Overview

The steps involved in our model

- **Paraphraser Model**: This will paraphrase the input statement into k statements. OpenAI's *text-davinci-003* will be used for this. After paraphrasing, there will be a total of $k + 1$ statements.

- **Quantity Tagging:** All the quantities in total $k + 1$ statements will be uniquely tagged. This is required so that the model can uniquely identify various quantities and generate results regardless of their specific values.

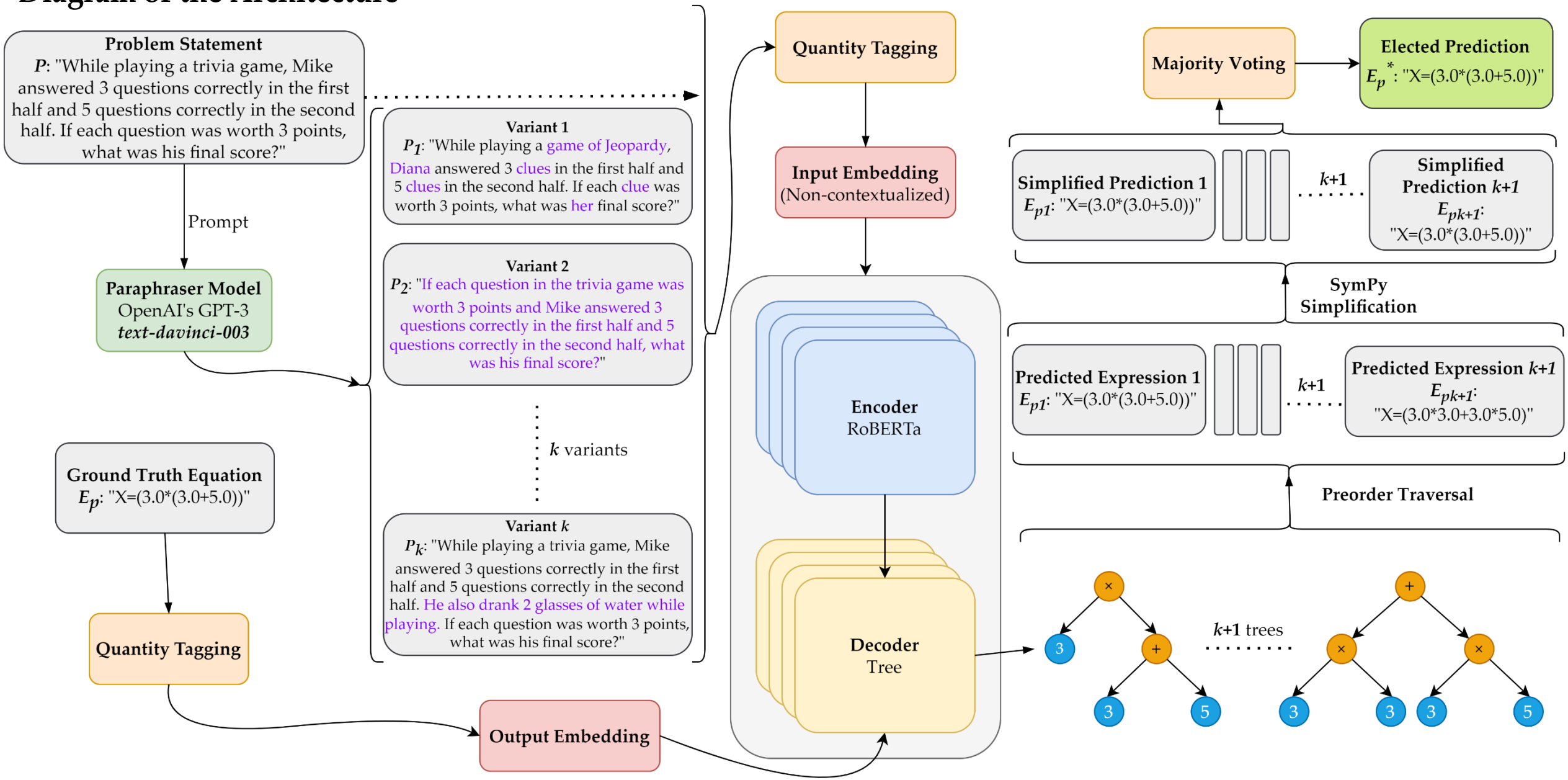
- **Encoder Model:** Only the encoder portion from RoBERTa Encoder-Decoder model will be used in our case. This will numerically encode all the generated variants of problem statement.

- **Tree Decoder Model:** The decoder portion will be designed in such a way that it returns a tree consisting of various quantities and operators. This tree can be traversed in pre-order manner to generate intended equations.

- **Majority Voting:** After simplifying each of the generated equations, they will be voted against each other and the equation that gets most number of votes will be regarded as the final answer.

Proposed Model

Diagram of the Architecture



Experimental Setup

A Summary of our Dataset Construction (*as of now)

- **PARAMAWPS (Paraphrased Math Word Problem Solving Repository)**
  - ✓ **Challenging** test-bed for judging robustness and efficacy of MWP solvers.
  - ✓ **New problems** generated from **selectively sampled seed problems** from **MAWPS**.
    - ➢ Variations in **names** of entities and objects
    - ➢ Variations in **sentence structure**
    - ➢ Variations by including **adversarial** and **irrelevant numerical information**
    - ➢ Variations by **inverting** problem statement
  - ✓ Total Problems — **4903**
  - ✓ Total Unique Equation Templates — **212**
  - ✓ Avg. operators per Problem — **1.971**
  - ✓ Avg. quantities per Problem — **2.88**
  - ✓ Avg. quantities per Equation — **2.94**
  - ✓ # of Problems with Constants — **738**

Experimental Setup

Types of Problems as Difficult Seeds

- Constant derived/implied from problem statement

Problem: Mary, Sam, Keith, and Alyssa each have 6 marbles. How many marbles do they have in all?
Equation: $x = 4 \times 6$

- Simple statement (1 direct quantity) yielding complex equation (requires external knowledge)

Problem: When the price of diesel rose by 10%, a user reduced his diesel consumption by the same amount. How much would his diesel bill change in terms of percentage?
Equation: $x = (1.0 - ((1.0 + (10.0 \times 0.01)) \times (1.0 - (10.0 \times 0.01)))) \times 100.0$

- Quadratic, known quantities in LHS

Problem: The Hudson River flows at a rate of 3 miles per hour. A patrol boat travels 60 miles upriver, and returns in a total time of 9 hours. What is the speed of the boat in still water?
Equation: $(60.0/(x - 3.0)) + (60.0/(3.0 + x)) = 9.0$

- Knowledge of mixtures required (0% fat vs 7% milk)

Problem: Lauren wants to mix 5 liters of 7% milk with skim-milk (0% fat) to produce a mixture of 2.9787% milk. How much skim-milk should Lauren add?
Equation: $x = (7.0 \times 0.01) \times 5.0/(2.9787 \times 0.01) - 5.0$

- And many more…

Tough even for GPT-3

Experimental Setup

Types of Linguistic Variations

- **Change phrase order:** Altering the order of appearance of phrases in the problem

Problem: There were originally 20817 houses in Lincoln County. During a housing boom, developers built 97741. How many houses are there now in Lincoln County?
Variant: How many houses are there in Lincoln County now, after developers built an additional 97741 during a housing boom, when there were originally 20817 houses?

- **Change object and entity names:** Altering the question by changing the way of conveying similar info.

Problem: While playing a trivia game, Mike answered 3 questions correct in the first half and 5 questions correct in the second half. If each question was worth 3 points, what was his final score?
Variant: While playing a game of Hangman, Emily guessed 3 letters correctly in the first half and 5 letters correctly in the second half. If each letter was worth 3 points, what was her final score?

- **Add unrelated information:** Including info. not required to solve the problem

Problem: A carpenter bought a piece of wood that was 8.9 centimeters long. Then he sawed 2.3 centimeters off the end. How long is the piece of wood now?
Variant: A carpenter bought a piece of wood that was 8.9 centimeters long. Then he sawed 2.3 centimeters off the end and sanded the wood for 20 minutes. How long is the piece of wood now?

- **Invert question:** Making a previously known quantity unknown, and vice versa

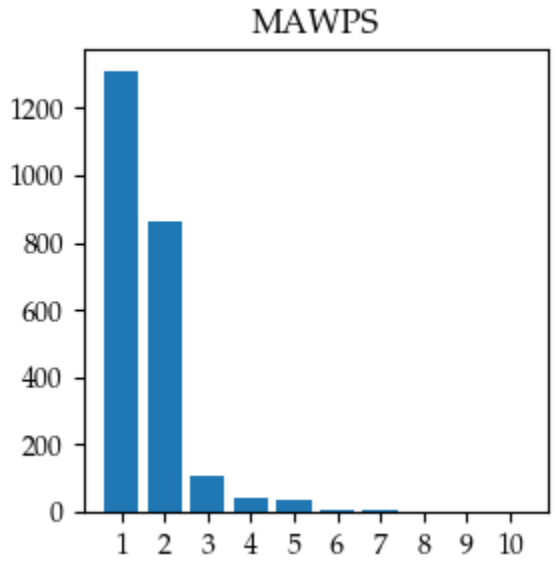
Problem: Mary bought 3 pizzas for $8 each. What was the total amount she paid for the 3 pizzas?
Variant: If Mary paid $24 for 3 pizzas, how much did she pay for each pizza?

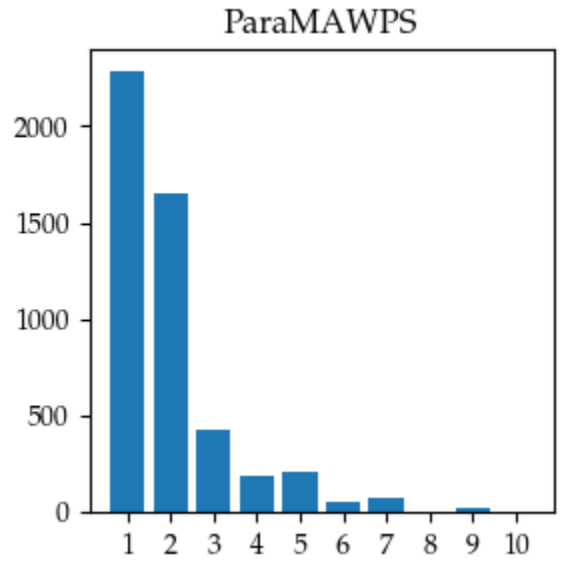
Experimental Setup

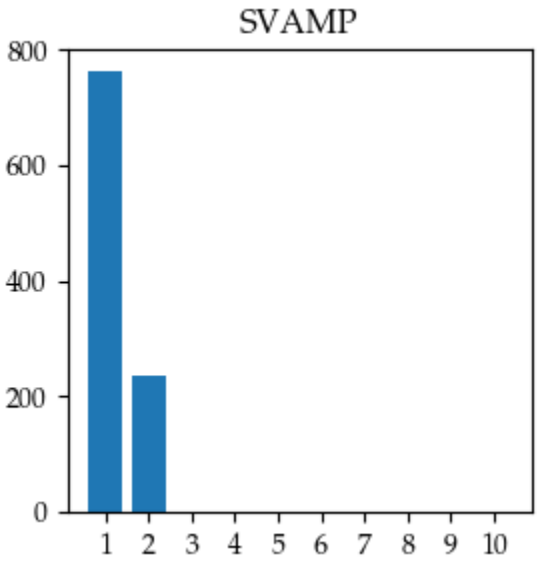
Dataset Stats Comparisons

Dataset	# of problems	# of unique templates	Avg. # of operators	Avg. # of quantities per problem	Avg. # of quantities per equation	# of problems with constants
SVAMP	1000	27	1.236	2.81	2.23	0
MAWPS	2373	159	1.606	2.57	2.59	185
ParaMAWPS*	**4903**	**212**	**1.971**	**2.88**	**2.94**	**738**

- More **variety** in types of problems, given by **higher unique template count**
- **Higher** amount of **complex problems**, given by **higher average operator count**
- **Similar** operator count **distribution** to MAWPS, giving **balance** between difficult problems and simple problems







Experimental Setup

Results

- Top-5 specific words with highest attention weights for each operator,

Operator	Addition (+)	Subtraction (-)	Multiplication (×)	Division (÷)
Words	all, together, more, total, adds	rest, take, took, more, and	every, per, total, product, percentage	Each, per, divide, percentage, share

- Performance Evaluation Table

Methods	SVAMP (%)		MAWPS (%)		PARAMAWPS (%)	
	Equ. Acc.	Ans. Acc.	Equ. Acc.	Ans. Acc.	Equ. Acc.	Ans. Acc.
GTS	25.6	29.1	82.6	83.5	-	-
Graph2Tree	31.6	35.0	83.7	85.6	-	-
RPKHS	-	-	89.8	-	-	-
RoBERTaDeductReasoner	**47.3**	-	**92.0**	-	-	-
Transformer	18.4	20.7	73.7	-	26.8	-
BERTGen	22.2	24.8	79.0	86.9	79.8	83.4
RoBERTaGen	27.9	30.3	80.8	88.4	70.5*	78.9*

Conclusion and Future Plan

What we plan to achieve

We propose the idea of a Math Word Problem solving framework that utilizes the paraphrased linguistic variations of problem texts to train a **RoBERTa-Tree** model that can solve the math word problem variants by giving out the solution expressions and then finalizing the predicted math expression using majority voting among the simplified candidate set expressions.

We also introduce a large-scale challenge dataset called **ParaMAWPS** that consists of paraphrased versions of selectively sampled datapoints from MAWPS along with inverse operation versions and adversarial versions of the MWPs, as a robust evaluation test-bed for MWP solver models.

Our plans for the future are:

- Complete the construction of the ParaMAWPS dataset and create a good challenge set.
- Implement our proposed architecture and test its efficacy.
- Experiment with harder problem text variations (*e.g.* grammatical errors) that contain multiple steps.
- Experiment with multi-variable equations and domain-knowledge problems.

References

Works Cited in This Presentation

[1] L. d. M. Daniel Selsam, "Imo grand challenge." https://imo-grand-challenge.github.io/.

[2] C. R. Fletcher, "Understanding and solving arithmetic word problems: A computer simulation," Behavior Research Methods, *Instruments, & Computers*, vol. 17, no. 5, pp. 565–571, 1985.

[3] Y. Bakman, "Robust understanding of word problems with extraneousinformation," arXiv preprint math/0701393, 2007.

[4] M. Yuhui, Z. Ying, C. Guangzuo, R. Yun, and H. Ronghuai, "Frame-based calculus of solving arithmetic multi-step addition and subtraction word problems," in *2010 Second International Workshop on Education Technology and Computer Science*, vol. 2, pp. 476–479, IEEE, 2010.

[5] N. Kushman, Y. Artzi, L. Zettlemoyer, and R. Barzilay, "Learning to automatically solve algebra word problems," in *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 271–281, 2014.

[6] S. Roy, T. Vieira, and D. Roth, "Reasoning about quantities in natural language," *Transactions of the Association for Computational Linguistics*, vol. 3, pp. 1–13, 2015.

[7] M. J. Hosseini, H. Hajishirzi, O. Etzioni, and N. Kushman, "Learning to solve arithmetic word problems with verb categorization.," in *EMNLP*, vol. 523533, Citeseer, 2014.

[8] L. Zhou, S. Dai, and L. Chen, "Learn to solve algebra word problems using quadratic programming," in *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pp. 817–822, 2015.

[9] A. Mitra and C. Baral, "Learning to use formulas to solve simple arithmetic problems," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2144–2153, 2016.

[10] C.-C. Liang, K.-Y. Hsu, C.-T. Huang, C.-M. Li, S.-Y. Miao, and K.-Y. Su, "A tag-based english math word problem solver with understanding, reasoning and explanation," in *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations*, pp. 67–71, 2016.

[11] C.-C. Liang, S.-H. Tsai, T.-Y. Chang, Y.-C. Lin, and K.-Y. Su, "A meaning-based English math word problem solver with understanding, reasoning and explanation," in Proceedings of COLING 2016, the *26th International Conference on Computational Linguistics: System Demonstrations*, (Osaka, Japan), pp. 151–155, The COLING 2016 Organizing Committee, Dec. 2016.

[12] S. Roy and D. Roth, "Solving general arithmetic word problems," *arXiv preprint arXiv:1608.01413*, 2016.

[13] S. Roy and D. Roth, "Illinois math solver: Math reasoning on the web," in *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations*, pp. 52–56, 2016.

References

Works Cited in This Presentation

[14] R. Koncel-Kedziorski, H. Hajishirzi, A. Sabharwal, O. Etzioni, and S. D. Ang, "Parsing algebraic word problems into equations," *Transactions of the Association for Computational Linguistics*, vol. 3, pp. 585–597, 2015.
[15] S. Roy, S. Upadhyay, and D. Roth, "Equation parsing: Mapping sentences to grounded equations," *arXiv preprint arXiv:1609.08824*, 2016.
[16] S. Roy and D. Roth, "Unit dependency graph and its application to arithmetic word problem solving," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 31, 2017.
[17] L. Wang, Y. Wang, D. Cai, D. Zhang, and X. Liu, "Translating a math word problem to an expression tree," *arXiv preprint arXiv:1811.05632*, 2018.
[18] T.-R. Chiang and Y.-N. Chen, "Semantically-aligned equation generation for solving and reasoning math word problems," *arXiv preprint arXiv:1811.00720*, 2018.
[19] S. Shi, Y. Wang, C.-Y. Lin, X. Liu, and Y. Rui, "Automatically solving number word problems by semantic parsing and reasoning," in *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pp. 1132–1142, 2015.
[20] Y. Zou and W. Lu, "Text2math: End-to-end parsing text into math expressions," *arXiv preprint arXiv:1910.06571*, 2019.
[21] D. Huang, S. Shi, C.-Y. Lin, J. Yin, and W.-Y. Ma, "How well do computers solve math word problems? large-scale dataset construction and evaluation," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 887–896, 2016.
[22] S. Upadhyay, M.-W. Chang, K.-W. Chang, and W.-t. Yih, "Learning from explicit and implicit supervision jointly for algebra word problems," in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pp. 297–306, 2016.
[23] D. Huang, S. Shi, C.-Y. Lin, and J. Yin, "Learning fine-grained expressions to solve math word problems," in *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pp. 805–814, 2017.
[24] A. Patel, S. Bhattamishra, and N. Goyal, "Are nlp models really able to solve simple math word problems?," *arXiv preprint arXiv:2103.07191*, 2021.
[25] Z. Xie and S. Sun, "A goal-driven tree-structured neural model for math word problems.," in *IJCAI*, pp. 5299–5305, 2019.
[26] L. Wang, Y. Wang, D. Cai, D. Zhang, and X. Liu, "Translating a math word problem to an expression tree," *arXiv preprint arXiv:1811.05632*, 2018.
[27] J. Zhang, L. Wang, R. K.-W. Lee, Y. Bin, Y. Wang, J. Shao, and E.-P. Lim, "Graph-to-tree learning for solving math word problems," *Association for Computational Linguistics*, 2020.
[28] J. Zhang, R. K.-W. Lee, E.-P. Lim, W. Qin, L. Wang, J. Shao, and Q. Sun, "Teacher-student networks with multiple decoders for solving math word problem," *IJCAI*, 2020.

References

Works Cited in This Presentation

[29] Z. Liang, J. Zhang, J. Shao, and X. Zhang, "Mwp-bert: A strong baseline for math word problems," *arXiv preprint arXiv:2107.13435*, 2021.

[30] W. Yu, Y. Wen, F. Zheng, and N. Xiao, "Improving math word problems with pre-trained knowledge and hierarchical reasoning," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 3384–3394, 2021.

[31] Z. Jie, J. Li, and W. Lu, "Learning to reason deductively: Math word problem solving as complex relation extraction," *arXiv preprint arXiv:2203.10316*, 2022

[32] R. Koncel-Kedziorski, S. Roy, A. Amini, N. Kushman, and H. Hajishirzi, "Mawps: A math word problem repository," in *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 1152–1157, 2016.

THANK YOU FOR LISTENING.